Climatological Lapse Rates and Freezing-Level Dynamics in Caucasus Mountain Network

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ABSTRACT

This research examines long-term temperature-elevation relationships and freezing-level variability using weather station data across an elevation gradient from the Caucasus and Andes mountain networks. Daily observations (raw datasets) were quality-controlled and aggregated into seasonal and annual climatologies. Linear regression between temperature and elevation was used to estimate lapse rates and freezing-level altitude. Results indicate stable but seasonally distinct lapse-rate behavior, with stronger gradients during the cool season and greater variability over time. Freezing-level altitude shows an overall increase over the years with significant interannual variability, highlighting potential implications for glacier melt and thermodynamically warm years. The analysis establishes a baseline framework for evaluating elevation-dependent climate change using observational station networks.

METHODS

Data Cleaning & Processing

Daily weather station observations were quality-controlled and standardized prior to analysis. Sentinel missing values (−9999.0) were converted to NaN, and rows containing no valid meteorological information were removed. Variables, units, and timestamps were harmonized across stations.

Data Aggregation

Cleaned daily data were aggregated into monthly and annual summaries.

Data Completeness Audit

A completeness assessment was performed to identify analyzable time periods. Months with insufficient observations were flagged as missing, and annual values were calculated only when adequate monthly coverage existed. Analyses were restricted to years with sufficient overlapping data across stations.

Lapse Rate Modeling

Regression: Fit **z = m × T + b** across stations

- **T** = that year's annual mean at each station
- **z** = station elevation
- **b** = freezing level (intercept at T = 0 °C)

Lapse rate (°C km−1^{-1}−1) was derived as: LR=−1000/m

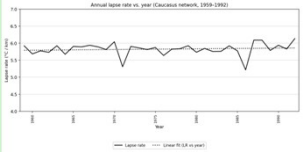
Freezing-Level Estimation

Freezing-level altitude was defined as the elevation corresponding to T=0°C, equal to the regression intercept (FL=b). Annual time series of lapse rate and freezing level were analyzed to evaluate variability and long-term trends.

RESULTS

Annual Lapse Rate

- **Period:** 1959–1992 inclusive
- **Per calendar year:** For each station, **annual mean temperature** was derived from the mean of monthly mean temperatures (tmean_c_month) over months with valid data, requiring **at least 10 valid months**.
- **LR_trend** = 0.00203 × year − 4.19 (°C / km)

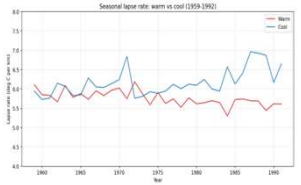


The trend line shows the lapse rate **increasing very slightly over 1959–1992, at a rate of about **+0.002 °C/km per year** (or roughly +0.07 °C/km per decade).
Annual regressions show consistently **high R^2 (>0.98) across all years.

Seasonal Lapse Rate

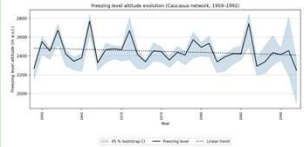
- **Period:** 1959–1992 inclusive
- **Per calendar year:** the **regional mean temperature** = average of monthly mean temperatures across all stations.
- **Warm season**: **top 4 months** by regional mean temperature
- **Cool season**: **bottom 4 months** by regional mean temperature
- **LR_trend** = 0.00203 × year − 4.19 (°C / km)

The **cool season consistently produces a steeper lapse rate than the warm season in most years Together, these two trends indicate a **widening warm/cool divergence** over the period, with a gap of **~0.2–0.5 °C/km in the 1960s growing to ~0.9–1.4 °C/km** by the late 1980s.



Freezing Level Timeseries

- **Period:** 1959–1992 inclusive
- A rising freezing level implies warmer conditions at elevation, accelerating melt; a falling freezing level implies the reverse.
- **Trend slope: −2.18m/year** (total decrease of ~74m over 34 years)



Maximum FL: 2768m
Minimum FL: 2245m
Mean FL: ~2453m
R^2: >0.98 in every year (confirms station network shows stable temp-elevation gradient)

SIGNIFICANCE

- **Long-term Steepening of Annual LR:** The weak positive trend of annual LR suggests a gradual strengthening of temperature with elevation → consistent with high elevations warming faster (Elevation Dependent Warming).
- **Strong Seasonal LR Contrast:** The cool season LR is consistently greater than the warm season LR across nearly all years → indicates stronger gradients in cool season due to inversions and cold-air pooling. This likely reflects enhanced wintertime atmospheric stability at low elevations, which is consistent with broader Northern Hemisphere circulation shifts and ENSO-related variability during this period. In contrast, warm season LR remain relatively stable due to convective mixing and moisture effects. This confirms that seasonality controls vertical temperature structure.
- **Freezing Level (FL) Behavior:** The mean freezing level is within station elevation range → confirms regression is physically grounded and mountain network captures the 0°C isotherm. Daytime shows higher FL and steeper LR while nighttime shows lower FL and weaker LR → diurnal control

Summary

The Caucasus network exhibits a stable vertical temperature structure with clear seasonal asymmetry. The observed cool season LR provide strong evidence for elevation-dependent warming, particularly in winter conditions. These results highlight seasonal atmospheric processes in shaping mountain climate change.



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